

DIY Ecological Momentary Assessment

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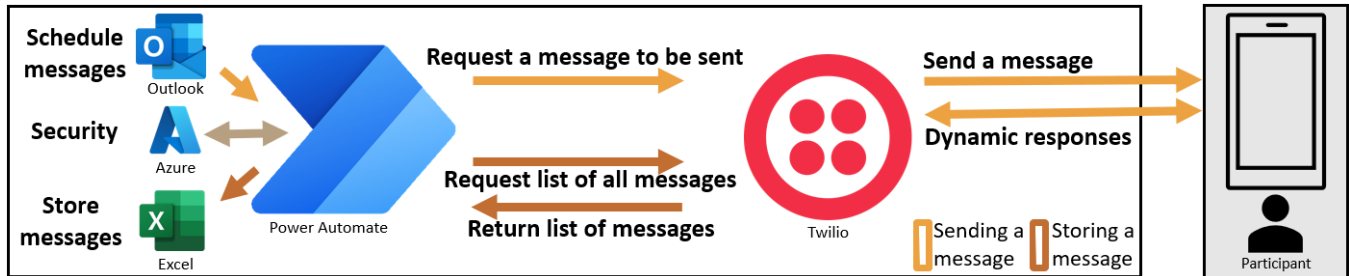


Figure 1: The off-the-shelf EMA system created by each participant by the end of the course.

ABSTRACT

Ecological Momentary Assessment (EMA) is an increasingly used methodology in HCI which captures an individual's experiences, typically digitally. The benefits of EMA include: reducing recall bias with 'in-the-moment' measurement; improving ecological validity with 'in-the-context' measurement and enabling exploration of the dynamic interplay of variables from repeated measurement. This course will teach participants about EMA by providing hands-on practical skill development in how to use off-the-shelf software to develop their own EMA system. At the end of the course, participants will be able to rapidly prototype and run custom EMA studies securely and at scale without the usual associated costs.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**.

KEYWORDS

Ecological Momentary Assessment, Experience Sampling, Rapid Prototyping

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1 BENEFITS

During the course participants will:

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- Learn more about EMA, a methodology that is being increasingly used in HCI to measure a wide range of subjective experiences, for example, mood, substance usage and binge eating [3][5].
- Understand the benefits of EMA for participant data collection, including reducing recall bias, increasing ecological validity and improving participant adherence [4].
- Create an EMA system using off-the-shelf tools, that are able to dynamically respond to study participants.
- Automate the scheduling of questions to participants while ensuring the secure transfer and storage of their responses using Microsoft technologies (Power Automate, Outlook calendar, Excel, Azure Key Vault).
- Understand the value of rapid prototyping in the context of EMA.

After the course participants will be able to apply the EMA system in their own projects and develop their own custom functionality using the tools they have learnt. Additionally, participants will be given takeaway materials that contain:

- Access to documentation that provides a detailed walk-through of how to create a customisable EMA system.
- The tools and documentation needed to justify the use of the system from a data protection and cost perspective to approval boards. This includes a data flow diagram which graphically represents the data policies of Twilio, Power Automate, Excel, Outlook and Azure.
- Twilio and Power Automate repositories, filled with different implementations of the system with corresponding documentation.
- Access to a repository of Masters students' work that extends the functionality detailed in the course.

2 INTENDED AUDIENCE(S)

This course is for students, researchers and practitioners who want to develop the knowledge and skills to rapidly prototype low-cost and secure EMA systems using off-the-shelf technologies.

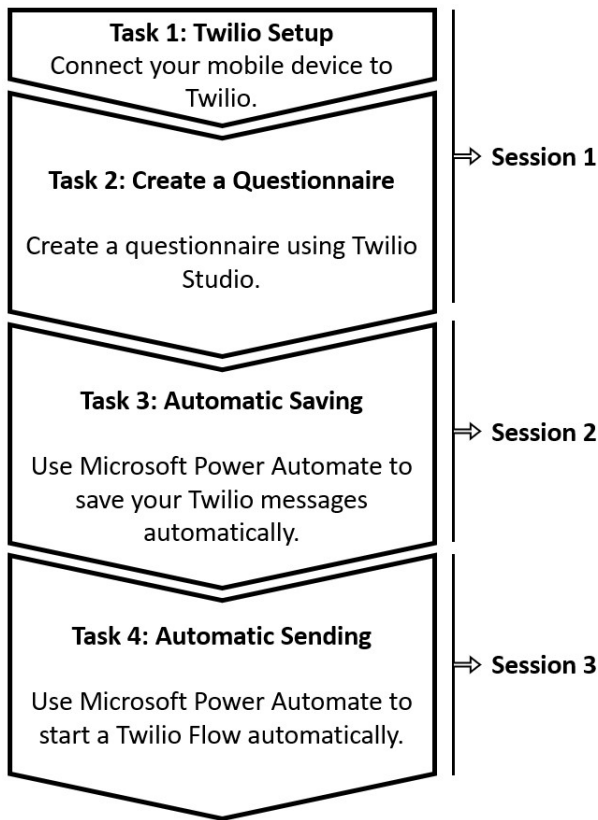


Figure 2: The breakdown of the set of tasks that will be used to guide participants to produce the final output in Figure 1 and how they map to the different sessions in the course.

3 PREREQUISITES

The course is open to anybody that has a basic understanding of how to use Excel e.g. how to make a table. No prior knowledge of EMA is required as an introduction to the methodology will be given in the course. Participants will need to bring a laptop to the session in order to engage with the material. All the software involved in this course can be accessed via web applications so no downloads are required before the session.

4 CONTENT

We begin with a brief introduction to EMA and how it is being used in HCI. We then highlight the benefits of being able to rapidly prototype a customisable, low-cost EMA system. Next, the participants will be introduced to the off-the-shelf technologies used to build the EMA system, using case studies to highlight how it has been used in small-scale projects. There will be a clear description of the benefits of this approach to developing EMA systems, including support for rapid prototyping; low cost; secure messaging and data storage; ease of use; potential to scale and easy customisation by a researcher. Participants will then learn the skills required to build their own EMA system by completing four tasks of increasing complexity as shown in Figure 2.

First Task. This task guides the participants through the use of **Twilio**, focusing on its visual interface which enables the messaging service to be used without any coding. Participants will be asked to connect their own phone, or an instructor-provided smartwatch, to Twilio. Once they have been able to connect their phone number (or the phone number of the instructor-provided device) to Twilio they can then test the basic message sending functionality. By setting up this basic messaging functionality, participants will understand Twilio’s visual interface features and how to navigate around the Twilio website. Additionally, participants will begin to see Twilio as a virtual phone number management system. This understanding is essential for utilising Twilio beyond its basic functionality.

Second Task. This task focuses on creating a one-question survey which is able to dynamically respond to a user-initiated conversation. To do this, participants will be introduced to **Twilio Studio**, which is the name of Twilio’s visual interface, and asked to create a **Flow**. A Flow involves combining different functional blocks (**Widgets**) to automate some communication functionality. The participants will be shown how to send a question, specifically “From 1 to 10 how are you feeling today?”, and then send a second message depending on whether the response was a low or high value. When previously running this course, we found that the understanding of Flows is consolidated by providing participants with laminated Widgets which they can use to paper prototype different functionality. They can then separate thinking about which Widgets provide the desired functionality from how to implement the Flow in Twilio Studio. Participants who are comfortable with creating a Flow in Twilio will be provided additional challenges that involve adding functionality to the basic Flow. The challenges will involve adding functionality to the basic Flow. These challenges were developed in response to feedback from previous course participants as well as researchers who have used our approach to build their own EMA system. By including the challenges as extensions, it allows differential learning and provides materials that participants can use when they are comfortable with using Flows.

Third Task. This task centres on **Power Automate**, a powerful platform for connecting many different applications. We focus on how it can link Twilio and Excel to provide secure data transfer and storage. To achieve this task, participants will be shown how to combine functional blocks to create a **Power Automate Flow**. These functional blocks will connect to an Excel document to store the data that has been transmitted during tasks one and two. We provide a detailed data flow diagram to facilitate participants’ understanding of how our approach follows best practices for data storage. The data flow diagram can be adapted by participants for their own projects in order to justify the use of the technologies from a data protection perspective to university and clinical bodies.

Fourth Task. In this task participants create a Microsoft Power Automate system that automates the sending of multiple messages to EMA participants. It is possible to do this using Twilio alone, but it would require some programming. We show participants how Power Automate can implement this functionality without the need for programming. This task will allow for differentiated learning. The simpler method uses **Outlook** to set up events manually to

start the Power Automate send message Flow. The more challenging method uses a combination of Twilio, Excel, Power Automate and Outlook to set up a schedule that takes into account study participants' availability and randomises the sending of messages. To achieve this task in Power Automate the workshop introduces **HTTP requests** and **API calls** and how these link Twilio and Power Automate. There is then an overview of data security practices and how to use **Azure Key Vault** to store **API keys** securely. In particular, they will be shown how to write an HTTP request to an API using Power Automate, a useful skill that enables connections to be made to many different applications. By the end of the workshop, the participants will have a system which is able to go through a list of study participants, send them messages, dynamically react to their responses and then store these messages securely in a separate Excel file.

5 PRACTICAL WORK

This course provides hands-on practical skill development in how to use off-the-shelf software to develop an EMA system, which participants have full control over, is low cost and implements high data protection standards. Twilio, is GDPR and HIPAA compliant as well as ISO accredited. Power Automate and Azure Key Vault both have ISO with GDPR and HIPAA compliance depending on how the user chooses to use these platforms. In our experience of running this course previously, all participants are able to develop a basic one-question EMA system after three and a half hours of tuition and hands-on activities. This basic EMA system can be used to run small pilot studies. We find that many participants that are more comfortable with the software are also able to add functionality to the basic system and improve its scalability. The course material shows participants how to create a multi-question EMA system which can dynamically respond to study participants' responses. The system supports message scheduling that can take account of study participants' availability, randomise the sending of messages and enable study participant-initiated self-reporting. Additionally, the more advanced EMA system can store all research data in a customisable format in Excel. If a participant does not implement all the more advanced functionality during the course then they can use the detailed resources we provide them with to develop their EMA system later.

6 INSTRUCTOR BACKGROUND

Sydney Charitos is a PhD student in Digital Health and Care at the University of Bristol and she is supported by an EPSRC doctoral training grant. Sydney created the original teaching materials and has used them to teach clinicians, researchers and students how to build EMA systems using this approach. With Jon Bird she has supervised Masters's students who have used the approach to rapidly build EMA systems to carry out research.

Dr Amberly Brigden is a Lecturer in Digital Health in the computer science department at the University of Bristol. She leads the University of Bristol's Ecological Momentary Assessment (EMA) Interest group, which spans the Faculty of Engineering and the Faculty of Health Sciences. She holds Research Grants focusing on EMA and is supervising EPSRC-funded PhD students to develop

EMA applications. Amberly teaches within the Faculty of Engineering and has been recognised for her teaching as a 2022 nominee for the Advanced HE Award for Teaching Excellence (CATE Award).

Dr Jon Bird is a Senior Lecturer in Digital health in the computer science department at the University of Bristol. He has taught university students for over 30 years and also has extensive experience in developing and running workshops and courses to teach technical skills, such as mobile app design and physical computing, to participants with non-technical backgrounds. Teaching technical skills is also a research interest and he has explored how to support troubleshooting when people are building systems that combine software and hardware [1][2]. He proposed the initial idea of using Twilio to develop EMA systems to Sydney who then fully developed the idea and found ways to use Power Automate, Excel and Azure to automate message scheduling and secure data storage. One of his current research interests is using EMA to better understand self-harm.

7 RESOURCES

As this approach to implementing an EMA system is completely novel the slides we provide are the only resource currently available. To gain more information about the software being used in the course, specific videos about Twilio and Power Automate have been identified which have a high level of relevance:

- Introduction to Twilio Studio: An introduction video to the Twilio Studio interface highlighting the purpose of the Studio tool to support rapid prototyping.
- Twilio Survey Tutorial guide: A guide which walks participants through how to make their own survey Flow with Twilio. This resource is similar to what will be created using Twilio Studio in the course, however, we will use Power Automate to schedule messages as this requires less technical expertise.
- Introductory Power Automate page: An introductory document on Power Automate, which guides individuals to create their own Flow.
- Power Automate Scheduled Flow Tutorial guide: A guide which focuses on teaching participants about scheduling triggers which means selecting specific dates and times to start a Power Automate Flow. Scheduled triggers are an important feature of the system described in the course.

8 ACCESSIBILITY

In terms of skill level accessibility, the workshop has been taught multiple times to individuals with widely varying technical expertise. As a result of feedback from previous course participants, the materials have been developed to support individuals who do not have programming skills. We have developed paper prototyping materials to facilitate participants' understanding of how to build Flows from functional blocks. The course materials have been developed to ensure that all technical terms are clearly explained. The sequence in which the technical materials are introduced has been carefully designed to facilitate learning. We provide clear instructions to guide participants through the workshop. The course supports differentiated learning, providing challenges for individuals who are progressing quickly and more hands-on time with

instructors for participants that need more support. In terms of accessibility to understand the information within the PowerPoint resources: images included in the PowerPoint will have alt text and key images will have detailed image descriptions; reading order is assigned on each slide; an audio transcript of the PowerPoint will be produced for hard of hearing individuals; PowerPoint slides will use a minimum of 12pt.

ACKNOWLEDGMENTS

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REFERENCES

- [1] Tracey Booth, Jon Bird, Simone Stumpf, and Sara Jones. 2019. Designing troubleshooting support cards for novice end-user developers of physical computing prototypes. In *International Symposium on End User Development*. Springer, 191–199.
- [2] Tracey Booth, Simone Stumpf, Jon Bird, and Sara Jones. 2016. Crossed wires: Investigating the problems of end-user developers in a physical computing task. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 3485–3497.
- [3] Soowon Kang, Cheul Young Park, Auk Kim, Narae Cha, and Uichin Lee. 2022. Understanding Emotion Changes in Mobile Experience Sampling. In *CHI Conference on Human Factors in Computing Systems*. 1–14.
- [4] Saul Shiffman, Arthur A Stone, and Michael R Hufford. 2008. Ecological momentary assessment. *Annu. Rev. Clin. Psychol.* 4 (2008), 1–32.
- [5] Niels Van Berkel, Denzil Ferreira, and Vassilis Kostakos. 2017. The experience sampling method on mobile devices. *ACM Computing Surveys (CSUR)* 50, 6 (2017), 1–40.